ENGINEERING DRAWING WITH AUTOCAD

UNIT 6: DEVELOPMENT OF SURFACES & AUTOCAD 3D OPERATIONS

Curriculum: Lovely Professional University | School of Computer Science & Engineering
Reference: Engineering Drawing with an Introduction to AutoCAD by Dhananjay Jolhe

>> 6.1 Concept and Industrial Applications of Surface Development

The development of a surface is the complete unfolding or unrolling of all outer bounding surfaces of a 3D solid onto a flat 2D plane without stretching, tearing, or distortion:
* Industrial Relevance: Essential for sheet metal fabrication, duct design (HVAC), pipe bends, chemical hoppers, aircraft fairings, boiler shells, and packaging design.
* True Length Principle: Every line and boundary edge drawn in a surface development must show its TRUE LENGTH (TL). Apparent foreshortened lengths from orthographic views cannot be used directly.

>> 6.2 Classification of Surface Development Methods

* 1. Parallel-Line Development: Used for solids having uniform parallel edges or generators (Prisms and Cylinders). The stretch-out line is a straight line perpendicular to the axis, with length equal to the perimeter of the base.
* 2. Radial-Line Development: Used for solids with converging edges or generators meeting at a single apex (Pyramids and Cones). The development is formed by radiating circular arcs centered at the apex, with radius equal to True Slant Height.
* 3. Triangulation Development: Used for transition pieces, hoppers, and irregular non-developable shapes. The surface is divided into a mesh of contiguous triangles whose true edge lengths are found and laid out consecutively.
* 4. Approximate Development: Used for double-curved surfaces that are mathematically non-developable (Spheres, paraboloids) by dividing into spherical zones (lunettes) or gore strips.

>> 6.3 Development of Truncated Solids and Cones

* Truncated Prism: Stretch-out line length = Perimeter (n x side). Vertical fold lines spaced by side width. Heights of cut points measured directly from elevation

view true lengths.
* Truncated Cylinder: Stretch-out line length = Circumference = pi x D. Divide circumference into 12 equal parts. Vertical generators erected and cut heights transferred horizontally.
* Truncated Cone: Sector of circle. Subtended angle theta = (Radius of base / True Slant Height) x 360 degrees = (R / L) x 360 deg. Divide sector into 12 equal angles and transfer cut generator points.
* Slant Edge True Length in Pyramids: If a pyramid slant edge is inclined to both HP and VP, rotate it parallel to the plane before measuring true length L.

>> 6.4 AutoCAD 3D Boolean Operations and Visual Styles

* Boolean Solid Modeling Commands:
- UNION (UNI): Merges two or more overlapping 3D solids or 2D regions into a single seamless continuous solid entity.
- SUBTRACT (SU): Removes the volume of selected tool solids from the designated base solid (Base Solid - Tool Solid).
- INTERSECT (IN): Calculates and retains only the common overlapping intersecting volume shared between two or more solids.
* 3D Navigation & Inspection:
- 3DORBIT (3DO): Dynamically rotates 3D model view in real time about target point.
- NAVISWHEEL: Unified steering wheel providing Pan, Zoom, Orbit, Rewind, and Look functions.
* Visual Styles (VS):
- 2D Wireframe: Basic vector lines and arcs representing edges.
- Conceptual: Smooth shaded surface with cool-to-warm color transitions based on face normals.
- Hidden: Displays solid geometry while suppressing hidden back lines.
- Realistic: Displays solid models with realistic materials, textures, and lighting.
- Shaded with Edges: Optimal engineering drafting mode displaying crisp black boundary edges over shaded faces.
* Mass Properties (MASSPROP): Calculates volume, mass, bounding box, center of gravity (centroid), moments of inertia, and radii of gyration for any 3D solid.

[PROBLEM 1: Parallel-Line Development of a Truncated Hexagonal Prism]

Problem Statement: A right regular hexagonal prism, base edge 30 mm and axis height 70 mm, rests on HP with one base edge perpendicular to VP. A cutting plane inclined at 45 degrees to HP cuts the prism axis at a height of 40 mm from the base. Draw the

development of the lateral surface of the truncated prism.
Step-by-Step Construction Procedure:
* 1. Front View & Section Plane:
- Draw Front View of hexagonal prism: Base line = 60 mm, Height = 70 mm.
- Axis passes at x = 30 mm. Mark cut center at height 40 mm on axis.
- Draw cutting plane at 45 degrees inclined to base passing through (30, 40).
- Label intersection points with vertical edges: 1'', 2'', 3'', 4'', 5'', 6''.
* 2. Stretch-Out Line Formulation:
- Stretch-out line length = 6 x side = 6 x 30 mm = 180 mm.
- Divide into 6 equal segments of 30 mm labeled A, B, C, D, E, F, A (seam at A).
- Erect vertical lines of height 70 mm at each division representing vertical edges.
* 3. Transferring True Cut Heights:
- Since vertical edges of the prism are perpendicular to HP, their elevation views show TRUE LENGTHS directly!
- Project cut point 1'' horizontally to edges A; point 2'' to edge B; point 3'' to edge C; point 4'' to edge D; point 5'' to edge E; point 6'' to edge F.
* 4. Boundary Profiling: Connect points 1, 2, 3, 4, 5, 6, 1 with straight lines (since prism faces are flat planar facets).

[PROBLEM 2: Radial-Line Development of a Truncated Right Circular Cone]

Problem Statement: A right circular cone of base diameter 60 mm and altitude 80 mm rests on its base on HP. It is cut by a section plane inclined at 30 degrees to HP, bisecting the cone axis. Draw the development of the remaining truncated cone surface.
Mathematical Derivation & Geometric Protocol:
* 1. Calculate True Slant Height (L):
- Radius R = 60 / 2 = 30 mm. Height H = 80 mm.
- Slant Height L = sqrt(R^2 + H^2) = sqrt(30^2 + 80^2) = sqrt(900 + 6400) = sqrt(7300) = 85.44 mm.
* 2. Calculate Sector Development Angle (theta):
- theta = (R / L) x 360 deg = (30 / 85.44) x 360 deg = 0.3511 x 360 = 126.4 degrees.
* 3. Radial Sector Layout:
- Center at Apex O, strike a circular arc of radius L = 85.44 mm.
- Measure sector angle theta = 126.4 degrees using protractor. Divide sector into 12 equal angles (10.53 deg each) representing 12 cone generators.
* 4. Finding True Cut Generator Lengths:
- Section plane bisects axis at height H/2 = 40 mm. Draw 30-deg cut plane.
- Only the extreme contour generators O-1'' and O-7'' in elevation show True Length.
- For intermediate generators (2, 3, 4, 5, 6), project intersection points HORIZONTALLY onto the extreme contour generator O-7'' to measure their true distances from apex O.

* 5. Plotting Cut Curve: Transfer these true radial distances from O along the corresponding radial generator lines in the sector. Connect points with a smooth French curve.

[PROBLEM 3: Development of a Square-to-Round Transition Duct]

Problem Statement: A sheet metal transition hopper connects a rectangular furnace flue (100 mm x 100 mm) at the base to a circular exhaust duct (dia 60 mm) at the top, vertical height 75 mm. Explain the Triangulation Development procedure.

Engineering Solution & Triangulation Procedure:

* 1. Symmetrical Decomposition: Due to 4-fold symmetry, divide the top circle into 12 equal parts (3 per quadrant: 1, 2, 3, 4). Each quadrant consists of a central flat triangle and two transitional scalene triangles connected to base corner A.
* 2. True Length Calculation via Right Triangles:
- Plan length = sqrt((X_top - X_base)^2 + (Y_top - Y_base)^2).
- True Length = sqrt((Plan length)^2 + (Vertical Height 75)^2).
* 3. Successive Triangle Construction: Lay down base edge AB = 100 mm. Strike arc from A with True Length A-1 and arc from B with True Length B-1 to locate apex 1. Continue constructing adjacent triangles A-1-2 and A-2-3 sequentially.

[PROBLEM 4: AutoCAD 3D Boolean Modeling & Mass Properties Analysis]

Problem Context: Create a hydraulic valve manifold body with intersecting bores using AutoCAD 3D Boolean tools and inspect mass properties.

AutoCAD Command Protocol:

* 1. BOX -> 0,0,0 -> Length 120 mm, Width 80 mm, Height 60 mm.
* 2. CYLINDER -> Center: 60,40,60 -> Radius: 20 mm -> Height: -60 mm (Vertical fluid chamber).
* 3. UCS -> 3P -> Align with front face (120 x 60 face).
* 4. CYLINDER -> Center: 60,30,0 -> Radius: 12 mm -> Height: -80 mm (Horizontal cross bore).
* 5. SUBTRACT Command: Select main Box -> Enter -> Select vertical and horizontal cylinders -> Enter.
* 6. FILLETEDGE -> Select all 12 outer sharp edges -> Radius: 3 mm. (Generates smooth cast fillets).
* 7. VISUALSTYLES -> Select "Shaded with Edges". Run 3DORBIT to inspect intersecting internal cavity.
* 8. MASSPROP Command: Select final manifold solid -> AutoCAD outputs exact Volume

(e.g., 478,214 mm^3), Centroid coordinates, and Moments of Inertia.